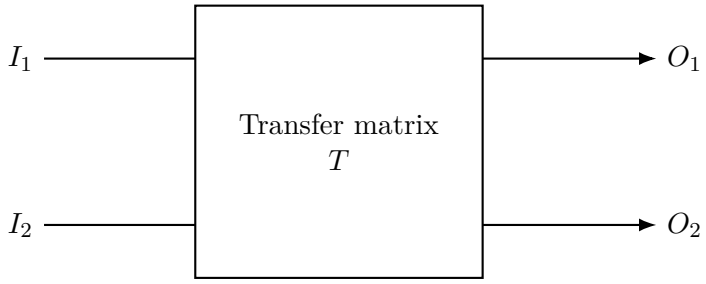
Understanding a Hamiltonian from an Input–Output Model

Quantum Mechanics for Engineers — Tutorial notes

1. A two-input, two-output system

Consider the simplest linear two-port-channel model. The box represents the system's transfer mapping; each output can depend on both inputs.



$$\begin{pmatrix} O_1 \\ O_2 \end{pmatrix} = \underbrace{\begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}}_{\text{input–output mapping}} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix}, \qquad \begin{cases} O_1 = T_{11} I_1 + T_{12} I_2, \\ O_2 = T_{21} I_1 + T_{22} I_2. \end{cases}$$

Read the indices: column j identifies the input channel; row i identifies the output channel. **Thus T_{12} is the contribution from input 2 to output 1, per unit input 2 with the other input set to zero. If $(I_1, I_2) = (0, 1)$, then $(O_1, O_2) = (T_{12}, T_{22})$.**

2. The same mapping idea for a Hamiltonian

For an orthonormal basis $\{|1\rangle, |2\rangle\}$, the Hamiltonian H is also a linear operator on a two-component vector:

$$H \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

Here the "output" $H|\psi\rangle$ is another vector, *not* a measurement outcome or a transmission probability. The engineering analogy identifies what each matrix element does; the physics of H comes from the time-evolution equation on the next page.

3. A Hamiltonian determines the rate of change

For a quantum state $|\psi(t)\rangle = c_1(t)|1\rangle + c_2(t)|2\rangle$, the Schrödinger equation is

$$i\hbar \frac{d}{dt}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

Equivalently,

$$\dot{c}_1 = -\frac{i}{\hbar}(H_{11}c_1 + H_{12}c_2), \qquad \dot{c}_2 = -\frac{i}{\hbar}(H_{21}c_1 + H_{22}c_2).$$

$$\begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} \longrightarrow \boxed{\begin{array}{c} -iH/\hbar \\ \text{rate generator} \end{array}} \longrightarrow \begin{pmatrix} \dot{c}_1(t) \\ \dot{c}_2(t) \end{pmatrix}$$

Meaning of the terms:

- H_{11} and H_{22} contribute to the phase evolution of their respective components (in this chosen basis).
- H_{12} multiplies c_2 in the equation for $\dot{c}_1$: coupling from component 2 into the rate of change of component 1.
- H_{21} multiplies c_1 in the equation for $\dot{c}_2$: the reverse coupling.

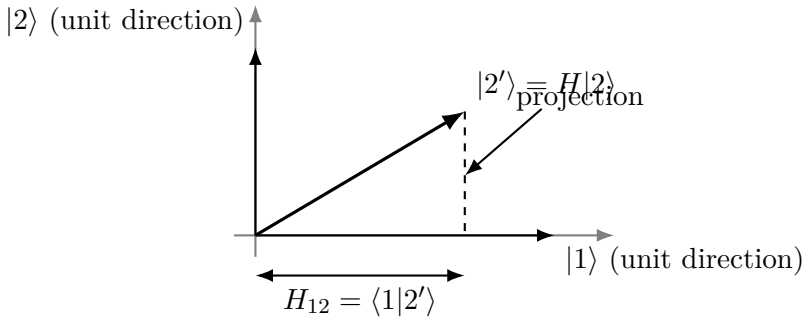
For a closed system, $H = H^\dagger$, so H_{11}, H_{22} are real and $H_{21} = H_{12}^*$. The matrix H does *not* directly map initial amplitudes to final amplitudes: for a time-independent Hamiltonian,

$$\boldsymbol{c}(t) = \underbrace{e^{-iHt/\hbar}}_{\text{finite-time evolution}} \boldsymbol{c}(0).$$

The Hamiltonian specifies the *instantaneous trend*; the exponential gives the evolution over time.

4. Why $H_{12} = \langle 1|H|2\rangle$: a geometric picture

Start with two orthonormal unit vectors $|1\rangle$ and $|2\rangle$. Apply H to the *input* $|2\rangle$, obtaining $|2'\rangle = H|2\rangle$. Then take its component along the *output direction* $|1\rangle$.



$$H|2\rangle = H_{12}|1\rangle + H_{22}|2\rangle, \qquad \underbrace{\langle 1|H|2\rangle}_{\text{input } 2 \to \text{ output component } 1} = H_{12}.$$

The diagram uses a real two-dimensional vector space for visualization. In a complex Hilbert space, H_{12} is generally a *complex coefficient*, extracted using the inner product $\langle 1|\cdot\rangle$; it need not be a real geometric length.

Takeaway: input column $\to$ linear operator $\to$ output component (row). For a Hamiltonian, these component couplings determine the *time derivatives* of quantum amplitudes.